ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 500 s	d/v.
2	**(c)** coastal low-lying areas	Flooding.
3	**(c)** evaporation	Heating.
4	**(c)** nitrogen fixation	Bacteria and lightning.
5	**(a)** using public transport and saving energy	Reduces emissions.
6	**(c)** 700 W per square metre	70 per cent absorbed.
7	**(c)** 1000 tonnes	20 x 50.
8	**(d)** run-off	Surface flow.
9	**(c)** converting N2 into nitrogen oxides	High energy.
10	**(b)** uses less electricity	Efficiency.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	t = 500 s (1); = 8.3 minutes (1).	1+1
12	Rhizobium in root nodules fix nitrogen (1); Soil gets nitrates (1).	1+1
13	Ice melts, lowering albedo (1); Warming is amplified (1).	1+1
14	Reflected = 300 W/m squared (1); Absorbed = 700 W/m squared (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Snow reflects 85 per cent (1); Forest and ocean reflect little, absorb a lot (1); Melting of snow increases warming (1).	1+1+1
16	Overuse of fertilisers (1); Burning fuels produces nitrogen oxides (1); Run-off leads to eutrophication (1).	1+1+1
17	Persist in the geosphere (1); Pollute the hydrosphere and harm marine life (1); Enter food chains (1).	1+1+1
	OR: Polar stratospheric clouds form in the cold (1); CFCs release chlorine (1); Chlorine destroys ozone in the sunlight of spring (1).	1+1+1
18	Absorb heat (1); Dissolve CO2 (1); Currents carry heat (1).	1+1+1
	OR: Photosynthesis releases O2 and absorbs CO2 (1); Respiration uses O2 and releases CO2 (1); Balance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Energy received = 2 kWh (1); Snow absorbs 0.3 kWh (1); Forest absorbs 1.8 kWh (1); Difference 1.5 kWh (1); Explains why forests are warmer (1).	1+1+1+1+1
	OR: Forest absorbs 600 tonnes (1); Cars emit 1440 tonnes (1); Forest offsets less than half (1); Need for emission cuts (1); Plant more trees (1).	1+1+1+1+1
20	Lightning and bacteria fix N2 (1); Nitrifying bacteria make nitrates (1); Plants absorb nitrates (1); Animals eat plants; decomposers return ammonia (1); Denitrifying bacteria return N2 (1).	1+1+1+1+1
	OR: Oceans 97 per cent (1); Ice 2 per cent, groundwater (1); Fluxes: evaporation, rain (1); Human: dams, pumping, pollution (1); Conservation (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs CO_2 and releases O_2 (plants release CO_2 only in respiration).
- Not converting units in rainwater problems: $\text{area} \times \text{depth}$ gives m^3, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 80 per cent	Fraction x 100.
2	**(c)** 6 kg	120 x 50 = 6000 g.
3	**(b)** groundwater	Infiltration.
4	**(a)** plants through roots	Absorption.
5	**(a)** reduces methane from landfills	Aerobic decay.
6	**(b)** forest	Absorbs more.
7	**(d)** 2.55 kWh	51 W x 50 h.
8	**(c)** condensation near the ground	Cooling.
9	**(b)** plants and other animals	Food chain.
10	**(c)** absorbing CO2 and providing shade	Positive effects.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Plants take up less CO2 in winter (1); More burning for heating (1).	1+1
12	Concrete and asphalt absorb heat (1); Fewer trees (1).	1+1
13	Ice reflects sunlight; water absorbs it (1); Lower albedo leads to more absorption (1).	1+1
14	It stays for hundreds of years in the atmosphere (1); It is removed slowly by plants and oceans (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sun's short waves enter (1); Earth re-radiates heat (1); Greenhouse gases trap part, balance sets temperature (1).	1+1+1
16	Contains most of the water vapour (1); Warm air rises and cools, forming clouds (1); Air mixing (1).	1+1+1
17	Force air to rise, causing rainfall on windward side (1); Store snow and ice (1); Feed rivers (1).	1+1+1
	OR: Power per square metre = 200 W (1); Area = 500/200 (1); = 2.5 square metres (1).	1+1+1
18	Annual: 300 x 12 x 0.8 = 2880 kg (1); Saving 20 per cent = 576 kg (1); New emission 2304 kg (1).	1+1+1
	OR: Store and filter water (1); Habitat for many species (1); Store carbon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Atmosphere CO2 (1); Photosynthesis to producers (1); Consumers and respiration (1); Decomposers and fossil fuels (1); Human burning and deforestation (1).	1+1+1+1+1
	OR: Early atmosphere had little oxygen (1); Cyanobacteria photosynthesis (1); Oxygen rose (1); Ozone layer formed (1); Allowed life on land (1).	1+1+1+1+1
20	Rise in average temperature (1); Melting glaciers (1); Sea-level rise (1); Extreme weather (1); Paris Agreement and net-zero goals (1).	1+1+1+1+1
	OR: Energy audit (1); Water saving (1); Waste segregation and composting (1); Plant trees (1); Measure by bills and weights (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs CO_2 and releases O_2 (plants release CO_2 only in respiration).
- Not converting units in rainwater problems: $\text{area} \times \text{depth}$ gives m^3, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 0.4	Fraction reflected.
2	**(a)** 2.04 kg of CO2	2.55 x 0.8.
3	**(a)** rises, cools and condenses	Condensation.
4	**(b)** nitrogen gas	Return to air.
5	**(a)** reduces plastic waste	LiFE habit.
6	**(d)** slanting rays	Angle.
7	**(d)** 0.4 mL	1000 x 0.0004.
8	**(d)** glaciers and ice caps	Frozen.
9	**(c)** pollution of water bodies	Run-off.
10	**(a)** saves standby energy	Saves power.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CO2 dissolves to form carbonic acid (1); pH about 5.6 (1).	1+1
12	Ground-level ozone is a pollutant harming lungs (1); Stratospheric ozone absorbs UV (1).	1+1
13	Releases CO2 (1); About 2.3 kg per litre (1).	1+1
14	Its parts interact through energy and matter (1); Change in one part affects others (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Oxides of sulphur and nitrogen dissolve in rain (1); Damage to marble and metals (1); Harm to soil and aquatic life (1).	1+1+1
16	Reflect sunlight, cooling (1); Trap heat at night, warming (1); Depends on the height and type (1).	1+1+1
17	Supports plants, stores water and carbon (1); Threats: erosion, pollution (1); Conservation: cover crops (1).	1+1+1
	OR: Nutrients are reused (1); Link living and non-living (1); Balance maintained (1).	1+1+1
18	Renewable: solar, wind (1); Non-renewable: coal, oil (1); Impact on environment (1).	1+1+1
	OR: Food: local and plant-rich (1); Transport: public transport, cycling (1); Energy: LED and star-rated appliances (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CFCs release chlorine (1); Ozone hole (1); Montreal Protocol bans (1); Recovery (1); Global co-operation and alternatives (1).	1+1+1+1+1
	OR: High sunshine in Rajasthan and Gujarat (1); Wind in Tamil Nadu (1); Intermittency (1); Storage in batteries (1); Grid improvement (1).	1+1+1+1+1
20	Equator: vertical rays (1); Mid-latitudes: moderate (1); Poles: slanting rays (1); Longer path through the atmosphere at the poles (1); Reflection by ice (1).	1+1+1+1+1
	OR: Geosphere releases lava and ash (1); Atmosphere: ash and gases cool the planet (1); Hydrosphere: acid rain (1); Biosphere: ecosystems destroyed then renewed (1); Soil becomes fertile (1).	1+1+1+1+1

Mistakes to avoid:

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- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
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